

1. The Platform at a Glance

PM CONTROL TOWER is an enterprise Project, Program and Portfolio Management platform built for top organizations that need one authoritative view of delivery. It carries a single dataset down the full management chain - portfolio, program, project, WBS, schedule, earned value, governance and leadership reporting - so every number a leader sees is computed from the same live operational data that teams update daily. There is no parallel reporting spreadsheet to reconcile and no month-end scramble to assemble board material.

The platform is deliberately executive-first. Role-based views give leaders a control tower of portfolio health, exceptions and decisions, while delivery teams work in the same system on tasks, timesheets and risks - no duplicate tools, no status bureaucracy. Realtime events keep every open screen current, and every create, update and decision is written to an immutable audit trail that auditors can inspect without engineering support.

104 REST API routes	57 data models	36 role-based views	8 governance roles
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2. The Module Map

Functionality is organized into eight module groups that mirror how an enterprise actually delivers work. Investment decisions happen at the portfolio and program level; planning happens once per project and is protected by baselines; control is exercised through engines and registers rather than meetings; execution stays lightweight for team members; and intelligence turns everything into reporting. The table below is the fastest way to navigate a live demo - each group maps directly to a section of the sidebar navigation.

Module group	Focus	Representative views
Portfolio	Investment and structure	Portfolios, Programs, Projects, Dashboard
Plan	Scope and schedule	Requirements, WBS, Schedule (CPM), Milestones, Baselines, Register
Control	Delivery assurance	EVM, RAID, Financials, Gates, Governance, Changes, Health
Execute	Team operations	Inbox, Timesheets, Planner, Resources
Intelligence	Reporting and insight	Leadership Control Tower, Reports, Analytics, Assistant
Connect	Integrations and automation	Integrations, Webhooks, Automations, Notifications

Module group	Focus	Representative views
Admin	Platform administration	Users, Roles, Audit, Templates, Settings
Project Workspace	Single-project depth	Overview, Scope, Schedule, RAID, EVM, Financials, Gates, Changes, Health, Audit tabs

Table 1: The eight module groups and their representative views.

3. Leadership Reporting and the Executive Control Tower

The Leadership Control Tower is the platform's differentiator. It answers the questions executives actually ask - are we within budget, what changed since the last report, which decisions are blocking delivery - from live data, with a strict honesty policy: any project with significant data gaps is classified GREY and can never present itself as green. Status is never self-declared; it is computed, versioned and comparable over time.

- **Exception-first feed** - critical risks, overdue decisions and due-soon milestones surface before everything else, so a leader reads the tower top-down in priority order.
- **Traceable auto-insights** - every sentence is derived from a named fact in the system; where evidence is missing the platform states INSUFFICIENT DATA instead of guessing.
- **Nothing-missed validation** - a 16-check audit of reporting completeness runs on every leadership pack, flagging exactly what a leader has not been told.
- **What-Changed engine** - every new report diffs automatically against the previous snapshot, making movement between reporting cycles explicit.
- **30/60/90 outlook** - upcoming milestones, decisions and financial exposure across the next three planning horizons.

Per-project Executive Status Reports follow a fixed leadership contract: a three-paragraph summary written from facts, a seven-dimension status assessment, and - most importantly - Leadership Actions in four fields: WHAT must be decided, WHO owns it, BY WHEN it is needed, and what happens IF NO DECISION is taken. Overdue decisions auto-escalate into critical action items on the next report run, so nothing stalls silently.

The one-click Leadership Pack assembles a complete, versioned board pack - control tower, exception feed, insights, and seventeen sub-reports spanning risks, issues, milestones, financials (with forecast trend against the previous snapshot), resources, schedule, dependencies, decisions, scope, deliverables, actions, quality and KPI realization. Packs are audited, scheduled (daily, weekly, monthly or quarterly) and comparable headline-to-headline; a dedicated Meeting Mode renders the pack as nine numbered sections for the leadership meeting itself.

20 sections per Leadership Pack	17 sub-reports, all computed	16 nothing-missed checks
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4. Decision Engines - EVM, CPM, Health and Governance

Beneath the reporting layer, six deterministic engines do the analytical work that organizations usually perform by hand. They run server-side on every relevant change, which means earned value, critical paths and health scores are always current - not a month old. This is what makes the control tower trustworthy: the numbers are produced by the same engines a PM would use, just continuously and for every project at once.

Engine	What it computes	What leadership gets
CPM schedule	ES/EF/LS/LF, float and critical path from live task dependencies	True critical path and slip impact per project
Earned Value	PV, EV, AC, CPI, SPI, EAC and variance against baseline	Objective cost and schedule health, forecast to completion
Health scoring	Weighted multi-factor RAG across scope, schedule, cost, risk and quality	One honest color per project, with contributing factors
Governance gates	Criteria-based stage gates with decision records	A defensible stage-decision history per project
Timesheet cascade	Submit, approve, reject, lock workflow feeding actuals	Actual effort and cost without manual collection
Portfolio rollup	Program and portfolio aggregation of progress, spend and health	Investment-level view that reconciles with every project

Table 2: The six decision engines and their executive outputs.

5. Security, Roles and Deployment

Security follows enterprise expectations: JWT authentication with bcrypt-hashed credentials, role-based access control across 32 permissions and 8 roles, rate limiting and schema validation on every API route, and a complete audit trail of who changed what and when. The 8 roles - EXECUTIVE, PMO_ADMIN, PORTFOLIO_MANAGER, PROGRAM_MANAGER, PROJECT_MANAGER, FINANCE, AUDITOR and TEAM_MEMBER - map to real organizational positions, and every view, report and API respects the matrix, down to 403s that auditors can verify.

Deployment is intentionally boring to operate. The reference path is Docker Compose - application plus realtime gateway plus an optional Caddy TLS edge - with the database on a persistent volume. Bare-metal Node and Bun paths, an offline dependency bundle for air-gapped environments, and a PostgreSQL upgrade route are all documented in the accompanying eleven-guide documentation suite, including backup, restore and runbook procedures.

Deployment path	Best fit	Notes
Docker Compose	Standard enterprise rollout	One command up; app + realtime + optional TLS profile; persistent volume for SQLite
Bare metal (Node or Bun)	Regulated hosts without containers	PM2 ecosystem file, nginx reverse proxy config and setup script included
Offline / air-gapped	Defense, government, secure data centers	Pre-built node_modules bundle with all native Linux x64 binaries; no internet needed
Cloud with managed Postgres	Scale-out or multi-region growth	Prisma schema ports to PostgreSQL; documented migration path

Table 3: Supported deployment paths.

The shipped reference dataset models a top organization: two portfolios, three programs and six projects with CPM schedules, baselines, RAID registers, budgets, forecasts, governance gates, decisions, actions and KPIs, plus twelve demo accounts across every role - including ceo@pmct.io (executive) and pmo@pmct.io (platform administrator). Everything shown below is the live product, not a mock-up. A suggested five-step demo path for the live session:

- **1.** Log in as ceo@pmct.io, open the Leadership Control Tower, read the exception feed.
- **2.** Drill into a red portfolio row's Executive Status Report and Leadership Actions block.
- **3.** Record the pending DEC-101 decision and watch the tower update and the exception clear.
- **4.** As pmo@pmct.io, generate a Leadership Pack and open Meeting Mode - nine numbered sections.
- **5.** Close the loop: liam@pmct.io files a timesheet, program@pmct.io approves - actuals hit EVM.